

Security Operations Center (SOC)

MVP

Software Architecture Design & Configuration Advice

Artur Kraskov
ICT & OL, Delta
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Contents

Contents.....	1
Introduction.....	1
Plan updates.....	1
Research summary.....	2
Technical Specification & Software Advice.....	2
Minimal Configuration for SOC MVP.....	2
Closed-source alternative.....	4
Trade-Offs: Open-Source vs. IBM QRadar SIEM.....	5
Why Choose an Open-Source SOC for SMEs or Startups?.....	6
When to Choose IBM QRadar SIEM?.....	6
Diagrams.....	7
C4 Level 1 - Software System.....	7
C4 Level 2 - Containers.....	8
C4 Level 3 - Components.....	8
Conclusion.....	9
Reference.....	9

Introduction

Within the scope of VINO AI for Cybersecurity project an idea of a Security Operations Center was researched and realized as MVP. This document provides technical stack description, advice and design of the SOC.

Plan updates

Current work expands previous plans and changes the direction from pure kill chain oriented solution to a more broad, distributed SOC with decision-making support capabilities [1].

Research summary

This work is based on research and extensive literature study. Full document is available as a PDF file via the link [2]. The research covered APT analysis, fractals in cybersecurity, MITRE ATT&CK framework, cyber kill chains (Lockheed, Extended, Unified), threat modelling and emulation with MITRE Caldera. It allowed us to uncover an existing problem with sophisticated cyber attack visualization, graph utilization. And a potential solution was found - QRadar offense chaining. Expanding on QRadar's SIEM interactive investigation platform an alternative was made with integrated Universal Matrix fractal graph. It offers consistent graph shape and potentially simplified or intuitive comparison.

Technical Specification & Software Advice

Minimal Configuration for SOC MVP

1. Basic Infrastructure

- **Hardware:**
 - 2 computers or virtual machines (host and target) with:
 - 8+ GB RAM
 - 100+ GB disk space
 - 2+ core CPU
 - 1 Mac OS computer or laptop for local dashboard deployment and acceleration
- **Network:**
 - A basic LAN or virtual network (can be on a single host using virtual machines or containers)
 - Described in the infrastructure document [3]

2. Core Components

Component	Purpose	Recommended Tools (Free/Open Source)
SIEM (Security Information & Event Management)	Centralized logging, alerting, and analysis	• Wazuh
Log Sources	Systems to send logs/events to SIEM & to provide logs to dashboard for analysis, visualization and investigation	• Windows Event Logs (Sysmon) • Snort IDS alert logs • Network traffic • Wazuh Alerts
IDS (Intrusion Detection System)	Detect suspicious traffic or activity on the network	• Snort
Threat Intelligence	Compare logs with known threats, get known mitigation strategies, descriptions	• MITRE ATT&CK Enterprise

Adversary Emulation	Deploy attack agents, emulate realistic cyber attacks, conduct red & blue exercises	• Caldera
Dashboard	Visualize and investigate alerts, accumulate SOC data from different sources in one place	• Streamlit • Plotly.js
Firewall	Secure ports responsible for SSH, TLS and C2	• UFW
VPN	Encrypted communication with Virtual Computing Environment	• Cisco Secure Client
Virtual Computing Environment	Virtual machines for remote deployment, hosting and research	• VMWare VSphere
AI	Decision making support, threat investigation	• Ollama server • Deepseek-r1 (reasoning model) • Gemma 3 (small and capable LLM)

The selected software stack for the SOC MVP is designed to balance cost-efficiency with high standards of availability, performance, security, and scalability. All chosen tools are open-source or free-tier industry standards, ensuring minimal cost without compromising capability.

- **Wazuh** serves as the SIEM platform, offering reliable real-time log aggregation, correlation, and alerting with high availability through clustering.
- **Snort** is employed as the IDS due to its proven performance in packet analysis and threat detection, making it suitable for both small-scale and scalable deployments.
- **MITRE ATT&CK** is integrated as a threat intelligence framework to enrich logs with actionable context, enhancing both detection and investigation quality.
- **MITRE Caldera** is a leading open-source automated adversary emulation solution. Referenced in research [4] and claimed to be used by the US government to test their systems.
- For log visualization and triage, **Streamlit** and **Plotly.js** provide lightweight, scalable dashboards that support rapid SOC analyst response. At the same time allow fast and reliable prototyping.
- **UFW** and **Cisco Secure Client** deliver foundational network security and encrypted remote access, ensuring secure communications within a virtualized infrastructure using **VMWare vSphere**.
- To assist in decision-making, **Ollama**, paired with **Deepseek-r1** and **Gemma 3:4b**, offers local AI inference for threat classification and investigation support, striking a balance between computational efficiency and analytical depth.

This architecture not only minimizes vendor lock-in but also supports modular growth, making it an optimal foundation for a production-ready SOC environment.

As an Integrated Development Environment (IDE) **Cursor AI Code Editor** was selected. With its free tier and accessible monthly subscription it provides an advanced coding AI companion with excellent code generation capabilities, different models, internet access and automation of various routine actions and processes. Based on VSCode it allows plugin integration and github version control. At the same time it is privacy oriented and SOC 2 certified. Great addition for SOC MVP framework - together with Streamlit it makes prototyping, integration of new components and development easier and faster.

More info: <https://www.cursor.com/>

Deploying the Streamlit SOC dashboard on **macOS with Apple Silicon (M1)** offers both **enhanced security** and **hardware-accelerated performance**. macOS provides a hardened operating environment, beginning with **System Integrity Protection (SIP)**, which restricts root-level modifications to critical system files—even for administrators—dramatically reducing the risk of persistent malware.

The **macOS firewall** is configured to **deny all incoming connections**, and **FileVault full-disk encryption** ensures that stored log data and sensitive configuration files remain secure at rest. Additional layers such as **admin/user account separation**, **Safari with hardened privacy settings**, and **third-party tools like Lulu and BlockBlock** enforce outbound firewall rules and monitor for unauthorized background processes or daemons, thereby minimizing attack surfaces and telemetry leakage.

On the performance side, Apple Silicon leverages the **Metal framework** for GPU-accelerated rendering, allowing **Streamlit and Plotly.js visualizations to render faster and more smoothly**, even with large datasets or dynamic graphs. This makes macOS not only a secure endpoint for dashboard hosting but also a **responsive, low-latency analysis node** within the SOC stack—ideal for on-the-go analysts or hybrid threat investigation workflows.

More info: <https://github.com/drduh/macOS-Security-and-Privacy-Guide>

Closed-source alternative

IBM QRadar SIEM is a mature, enterprise-grade SIEM platform that excels in core SOC domains—availability, performance, security, and scalability—while minimizing operational risks through its proprietary support and polished UI. Its architecture is optimized for high availability via clustering, and its threading and event-processing layers handle high-throughput (EPS/FPM) deployments with minimal latency.

More info: <https://www.ibm.com/qradar>

While pricing is usage-based—typically measured in events-per-second or managed virtual servers—alternatives exist for endpoint-based models. For example, at an entry tier supporting ~100 endpoints, QRadar costs approximately € 390 - € 468 /month, which, combined with 24×7 IBM support, delivers strong ROI through automated threat correlation and AI-powered prioritization.

Source: <https://www.ibm.com/products/qradar-siem/pricing#priceestimator>

Though not open-source, it offers seamless integration with MITRE ATT&CK, IBM X-Force threat intel, and over 700 prebuilt connectors, reducing development and maintenance overhead. This closed ecosystem can simplify SOC MVP roadmap—trading some flexibility for proven reliability and vendor accountability, ideal when uptime and analyst productivity are KPIs.

Trade-Offs: Open-Source vs. IBM QRadar SIEM

Factor	Open-Source SOC (e.g., Wazuh + Snort + Streamlit)	IBM QRadar SIEM (Closed-Source)
Cost	✓ Free or low-cost; ideal for MVPs or budget-limited setups	✗ € 390 - € 468/month for 100 endpoints and scales up quickly; cost includes support, updates
Customization	✓ Fully customizable; complete control over rules, integrations, dashboards	✗ Limited customization; black-box internals, relies on IBM's update cycles
Vendor Lock-In	✓ Avoided; full code access, no contractual ties	✗ High; migrating away can be complex and expensive
Ease of Setup	✗ High manual effort; multiple tools, formats, and integrations to manage	✓ Out-of-the-box SIEM; quick onboarding with IBM-provided integrations
Maintenance	✗ High; manual updates, patching, log tuning required	✓ IBM handles updates, patches, and security advisories
Scalability	⚠ Possible but requires effort; needs careful resource planning	✓ Scales easily with enterprise-grade architecture and licensing options
Security	✓ Transparent; code is inspectable, community-audited	⚠ Trusted but opaque; closed code base, must trust vendor's handling

Compliance	⚠ Manual setup of policies/log retention standards	✅ Preconfigured compliance packs (PCI, GDPR, HIPAA, etc.)
Support	❌ Community-based or self-managed	✅ 24/7 enterprise support and SLAs
Integration	⚠ Requires engineering effort (e.g., log parsing, dashboard UX)	✅ 700+ IBM-supported integrations (cloud, EDR, firewalls, etc.)
AI/Analytics	⚠ Add-ons required; less mature	✅ Built-in anomaly detection, AI-based prioritization, threat intelligence

Why Choose an Open-Source SOC for SMEs or Startups?

For **SMEs and startups**, a custom open-source SOC offers a **cost-effective and flexible** approach to building foundational security capabilities without the financial burden of commercial licensing. Tools like **Wazuh (SIEM)**, **Snort (IDS)**, and **Streamlit/Plotly.js (dashboards)** are widely used in the industry and provide **enterprise-grade features at no licensing cost**. This stack gives organizations **full control** over data flow, alert logic, and integrations, enabling them to tailor the system to their specific operational context or compliance needs.

While the open-source route demands **technical expertise** and some initial setup effort, it allows SMEs to:

- **Avoid vendor lock-in**, with the freedom to swap components as they grow.
- **Build institutional knowledge**, enhancing their long-term security maturity.
- **Scale gradually**, adding new modules (like AI, threat intel, or SOAR) without proprietary constraints.
- Leverage **community-driven innovation**, frequent updates, and transparent security.

However, these benefits come with trade-offs. SMEs must be ready to **invest time in configuration, log tuning, and maintenance**, or consider outsourcing parts of the management. Still, for organizations that need **affordable, auditable, and customizable SOC capabilities**, open-source stacks provide a **strategic foundation** that can evolve over time—making them particularly valid during early or transitional stages of growth.

When to Choose IBM QRadar SIEM?

IBM QRadar SIEM is the right choice for organizations that prioritize **rapid deployment, enterprise support, and long-term scalability** over full customization. It is particularly suitable for **SMEs scaling into regulated or high-risk industries** (e.g., finance, healthcare,

manufacturing) where **compliance, availability, and security guarantees** are non-negotiable. QRadar offers a **predictable, all-in-one platform** that reduces the need for in-house engineering effort.

Key advantages include:

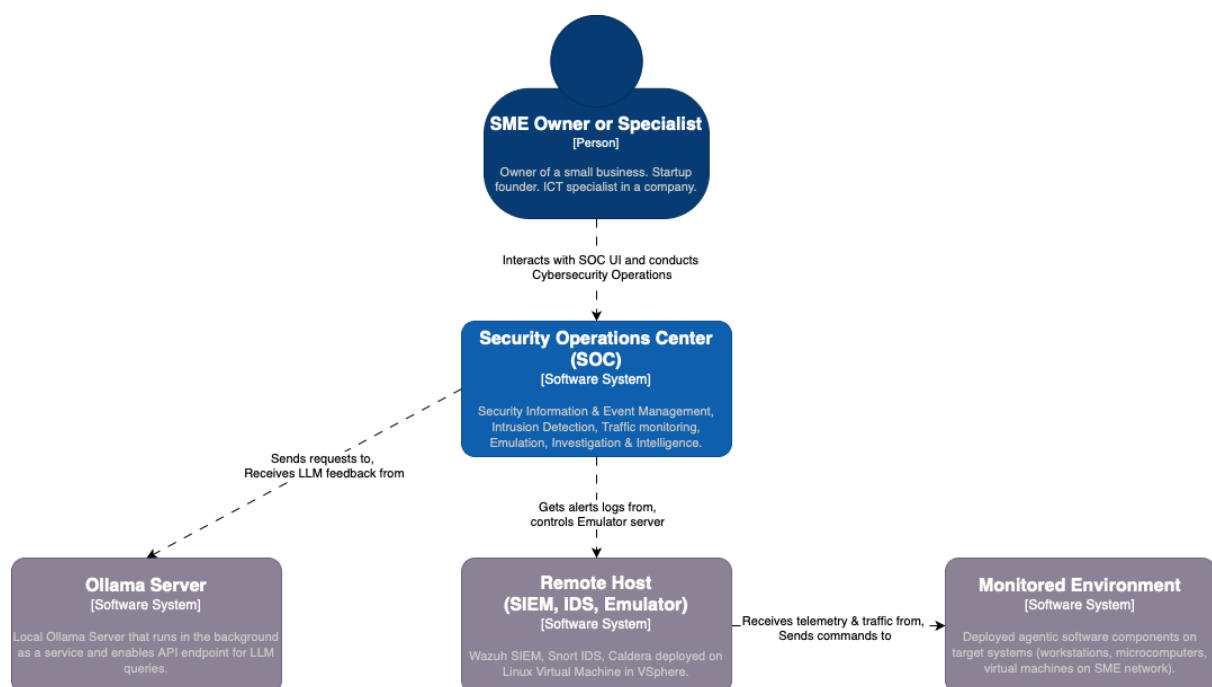
- **Out-of-the-box integrations** with 700+ data sources (cloud, firewall, EDR, etc.), significantly reducing setup time.
- Built-in support for **MITRE ATT&CK, IBM X-Force threat intelligence**, and **predefined compliance packs**(e.g., PCI DSS, GDPR, HIPAA).
- **24/7 enterprise support**, automated patching, and threat detection updates from IBM's security teams.
- **High scalability and performance** with minimal downtime due to its clustered architecture and EPS-based licensing model.
- Advanced features such as **AI-powered threat prioritization, user behavior analytics (UBA)**, and centralized forensic investigation tools.

Choose IBM QRadar when an organization needs **speed, stability, and accountability** from a SOC platform with proven results and trusted vendor backing. It is especially effective when internal teams are limited or when **audit readiness and SLA-backed uptime** are critical business requirements.

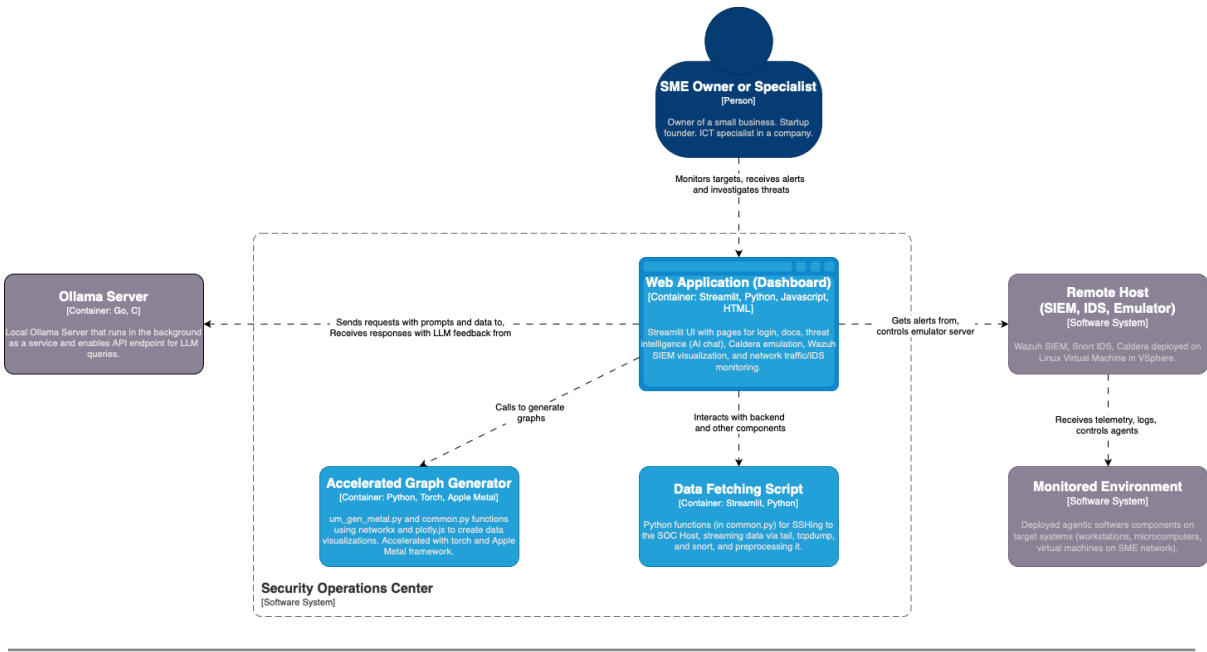
Diagrams

High resolution file:  [5]

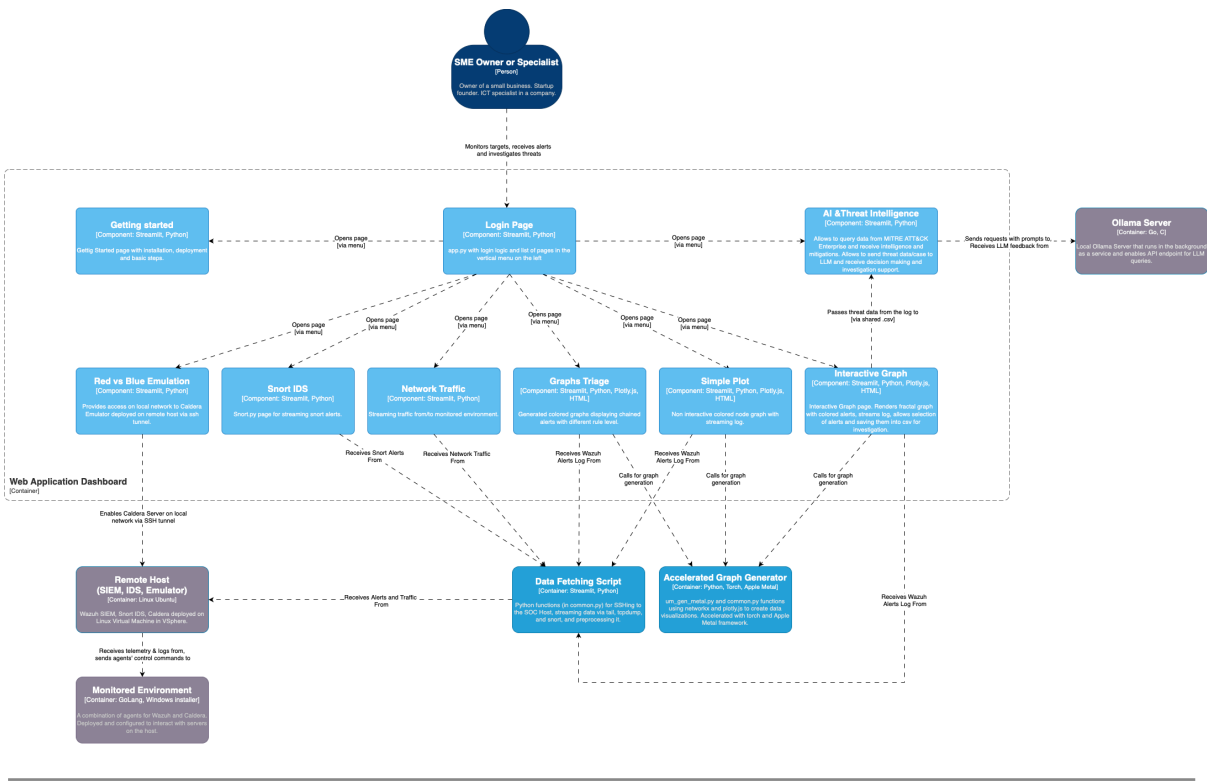
C4 Level 1 - Software System



C4 Level 2 - Containers



C4 Level 3 - Components



Conclusion

Proposed software architecture of the minimal SOC configuration allows all basic needs of emerging cybersecurity specialists or analysts. It can facilitate monitoring, analytics, live visuals for intuitive decision making support and even incident responding. By using this solution as a starting platform a potential SME stakeholder can build foundational knowledge and proceed with confidence and a clear roadmap to scalability and expansion.

In order to improve the solution in the future a necessary shift to a more low level visualization framework is advised. C/C++ Kibana or Shaders can be a potential direction to decrease latency. Yet, it can introduce a trade off with ease of development. Current solution is intended for users that want quick and measurable results, ability to control and audit everything. As well as customize and upgrade the default configuration.

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